

PRODUCT	MEMBER #	MIL	GAGE	COATING
Smart Joist	800J300-54SJ	54	16	G90

800J300-54SJ Product Submittal

PHYSICAL PROPERTIES

Web Depth	8"
Flange	3"
Return (lip)	0.625"
Approx. Weight (lb/ft)	2.93
Area (in^2)	0.8632
Design Thickness	0.0566
Minimum Thickness	0.0538
Yield Strength	50 KSI
Coating	G90

PRODUCT VIEW



GROSS PROPERTIES

Gross section moment of inertia about X-X axis (Ix)	8.8130
Section modulus about X-X axis (Sx)	2.2032
Radius of gyration about X-X axis (Rx)	3.1953
Gross section moment of inertia about Y-Y axis (Iy)	1.0523
Radius of gyration about Y-Y axis (Ry)	1.1042

EFFECTIVE / NET PROPERTIES

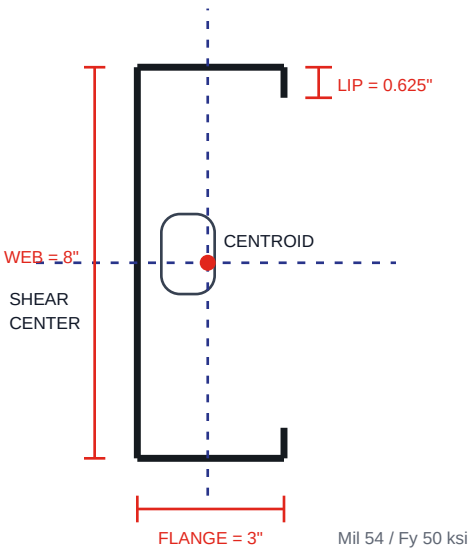
Net area at punchout (in^2)	0.6226
Net moment of inertia X-axis (Ix-net)	8.4509
Net section modulus X-axis (Sx-net)	2.1127
Net radius of gyration X-axis (Rx-net)	3.6842
Net moment of inertia Y-axis (Iy-net)	0.8841
Net radius of gyration Y-axis (Ry-net)	1.1917

TORSIONAL PROPERTIES

St. Venant torsional constant (Jx1000)	0.9217
Torsional warping constant (Cw)	Source-limited
Polar radius of gyration (Ro)	Source-limited

CAD SECTION

schematic dimensions



NOTES

Code Approvals & Performance Standards

- AISI S100-16 (2020) w/S2-20 North American Specification for the Design of Cold-Formed Steel Structural Members.
- AISI S240-20 North American Standard for Cold-Formed Steel Structural Framing.
 - o Compliant to ASTM C955; IBC replaced ASTM C955 with AISI S200 in IBC 2015 and AISI S240 in IBC 2018.
 - o Section A3 Material - Chemical and mechanical requirements (referencing ASTM A1003/A1003M).
 - o Section A4 Corrosion Protection (referencing ASTM A653/A653M).
 - o Section A5 Products - Thickness, shapes, tolerances, identification.
 - o Section C Installation (referencing ASTM C1007).
- AISI S202-20 Code of Standard Practice for Cold-Formed Steel Structural Framing.
 - o Section F3 Delivery, Handling, and Storage of Materials.
- IBC 2024 International Building Code.
- ICC-ES ESR-5837 Structural Studs and Track.
 - o ESR-5837 Catalog Refined Metal Technical Design Guide (10/1/2025).
- SDS for ASTM A1003 Steel Framing Products for interior framing, exterior framing, and clips/accessories.
- The technical content of this literature is effective 08/26/25 and supersedes all previous information.

DISCLAIMER

All data, details, and specifications included herein are intended as a general guide for using Refined Metal products. These products should not be used in design or construction without evaluation by a qualified engineer or architect to determine suitability for a specific use. Refined Metal assumes no liability for failure resulting from use or misapplication of computations, details, or specifications contained herein. Refined Metal assumes no liability for damages resulting from improper application or installation of these products.

DERIVED SMART JOIST SPAN TABLES

Preliminary single-span estimates in feet. Verify with manufacturer span tables or sealed engineering before use.

	15 psf dead load, 40 psf live load		15 psf dead load, 60 psf live load		15 psf dead load, 125 psf live load	
Spacing OC	L/360	L/480	L/360	L/480	L/360	L/480
12	18.5	16.8	16.8	15.3	13.7	12.5
16	16.9	15.4	15.3	13.9	12.5	11.3
24	14.8	13.5	13.4	12.2	10.9	9.9

	40 psf dead load, 40 psf live load		40 psf dead load, 60 psf live load		40 psf dead load, 125 psf live load	
Spacing OC	L/360	L/480	L/360	L/480	L/360	L/480
12	16.5	14.9	15.3	13.9	13.0	11.8
16	15.0	13.6	13.9	12.7	11.8	10.8
24	13.1	11.9	12.2	11.1	10.4	9.4

NOTES

Code Approvals & Performance Standards

- AISI S100-16 (2020) w/S2-20 North American Specification for the Design of Cold-Formed Steel Structural Members.
- AISI S240-20 North American Standard for Cold-Formed Steel Structural Framing.
 - o Compliant to ASTM C955; IBC replaced ASTM C955 with AISI S200 in IBC 2015 and AISI S240 in IBC 2018.
 - o Section A3 Material - Chemical and mechanical requirements (referencing ASTM A1003/A1003M).
 - o Section A4 Corrosion Protection (referencing ASTM A653/A653M).
 - o Section A5 Products - Thickness, shapes, tolerances, identification.
 - o Section C Installation (referencing ASTM C1007).
- AISI S202-20 Code of Standard Practice for Cold-Formed Steel Structural Framing.
 - o Section F3 Delivery, Handling, and Storage of Materials.
- IBC 2024 International Building Code.
- ICC-ES ESR-5837 Structural Studs and Track.
 - o ESR-5837 Catalog Refined Metal Technical Design Guide (10/1/2025).
- SDS for ASTM A1003 Steel Framing Products for interior framing, exterior framing, and clips/accessories.
- The technical content of this literature is effective 08/26/25 and supersedes all previous information.

DISCLAIMER

All data, details, and specifications included herein are intended as a general guide for using Refined Metal products. These products should not be used in design or construction without evaluation by a qualified engineer or architect to determine suitability for a specific use. Refined Metal assumes no liability for failure resulting from use or misapplication of computations, details, or specifications contained herein. Refined Metal assumes no liability for damages resulting from improper application or installation of these products.